

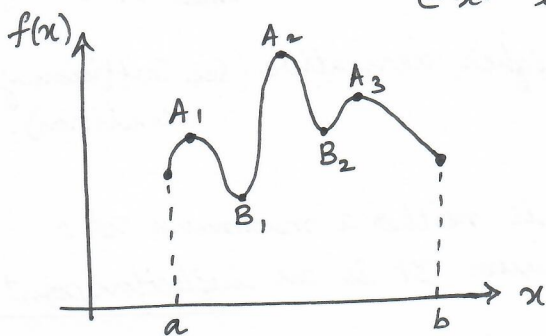
Chapter 2: Unconstrained Optimization① Single-Variable Optimization:

Local minimum: A function of one variable $f(x)$ is said to have a local minimum at $x=x^*$ if $f(x^*) \leq f(x^*+h)$ for all sufficiently small positive and negative values of h .

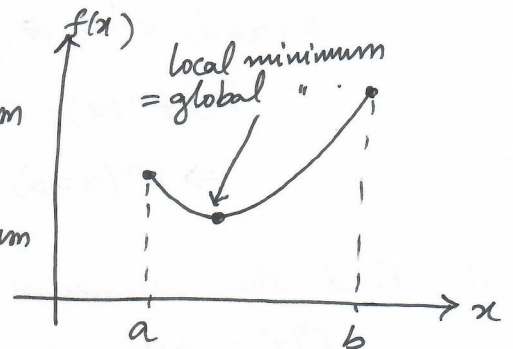
Local maximum: x^* would be local maximum if $f(x^*) \geq f(x^*+h)$ for all values of h sufficiently close to zero.

Global minimum: @ x^* if $f(x^*) \leq f(x) \forall x$ (not just x close x^*) in the domain over which $f(x)$ is defined.

Global maximum: @ x^* if $f(x^*) \geq f(x) \forall x$ in the domain.



$A_1, A_2, A_3 = \text{local maxima}$
 $A_2 = \text{Global maximum}$
 $B_1, B_2 = \text{local minima}$
 $B_1 = \text{Global minimum}$



- A single-variable optimization problem is one in which the value of $x=x^*$ is to be found in the interval $[a, b]$ such that x^* minimizes $f(x)$.

- Necessary condition for a point x^* to be a minimum:

If a function $f(x)$ is defined in the interval $a \leq x \leq b$ and has a relative ^(local) minimum at $x=x^*$, where $a < x^* < b$, and if the derivative $\frac{df(x)}{dx} = f'(x)$ exists as a finite number at $x=x^*$,

then $\boxed{f'(x^*) = 0}$ i.e. $\boxed{\left. \frac{df(x)}{dx} \right|_{x=x^*} = 0}$

- Sufficient condition for a point x^* to be a minimum.

Let $f'(x^*) = f''(x^*) = \dots = f^{(n-1)}(x^*) = 0$, but $f^{(n)}(x^*) \neq 0$. Then $f(x^*)$ is

- a minimum value of $f(x)$ if $f^{(n)}(x^*) > 0$ and n is even
- a maximum value of $f(x)$ if $f^{(n)}(x^*) < 0$ and n is even
- neither a maximum nor a minimum if n is odd.

Note: A "necessary condition" is required for a point to be a minimum, but it is not sufficient, (i.e. further conditions ("sufficient condition") are required for that) (E.g. $f'(x^*) = 0$ need not always mean min. or max.)

- Examples:

(i) Determine the maximum and minimum values of the function

$$f(x) = 12x^5 - 45x^4 + 40x^3 + 5$$

Solution:

Solution: $f'(x) = 60(x^4 - 3x^3 + 2x^2) = 60x^2(x-1)(x-2)$

so $f'(x) = 0 \Rightarrow x = 0$ or 1 or 2 .

$$f''(x) = 60(4x^3 - 9x^2 + 4x)$$

$$f''(x=1) = -60 < 0 \Rightarrow x=1 \text{ is a local } \text{maximum} \Rightarrow f_{\max} = f(1) = 12$$

$$f''(x=2) = 240 > 0 \Rightarrow x=2 \text{ is a local minimum} \Rightarrow f_{\min} = f(2) = -11$$

$f''(x=0) = 0 \Rightarrow$ need to find next higher derivative (see sufficiency condition).

$$\text{So, } f'''(x) = 60(12x^2 - 18x + 4)$$

$\Rightarrow f'''(x=0) = 240 \neq 0 \Rightarrow x=0$ is neither a maximum nor a minimum. It is an inflection point.

(ii) Find the minimum of $f(x) = x^4$, using sufficiency conditions.

$$f'(x) = 4x^3, \text{ so } f'(x) = 0 \Rightarrow x^* = 0$$

$$f''(x) = 12x^2 \Rightarrow f''(x^*) = 0 \Rightarrow \text{need to find higher derivative.}$$

$$f'''(x) = 24x \Rightarrow f'''(x^*) = 0 \Rightarrow \quad " \quad " \quad "$$

$$f''(x) = 24 \Rightarrow f''(x^*) = 24 > 0 \Rightarrow x_{=0}^* \text{ is a minimum.}$$

Recall: $f^{(n-1)}(x^*) = 0$ & $f^{(n)}(x^*) \neq 0$, if n is even then $f^n(x^*) > 0$ (or a minimizer).
In above e.g. $n=4$ min if $f^n(x^*) > 0$.

Exercise:

Taylor series expansion of $f(x)$ about $x = x^*$ (i.e. $h = x - x^* = \Delta x$)

$$f(x) = f(x^*) + \frac{df}{dx}\bigg|_{x^*} (x-x^*) + \frac{1}{2} \frac{d^2f}{dx^2}\bigg|_{x^*} (x-x^*)^2 + \underline{O(3)}$$

$$(2) \quad f(x) = f(x^*) + f'(x^*) \cdot \Delta x^* + \frac{1}{2} \cdot f''(x^*) \cdot (\Delta x^*)^2 + \underbrace{O((\Delta x^*)^3)}_{\text{higher order terms}}$$

For $\Delta x^* \rightarrow 0$ (very small), see how the necessary & sufficient conditions make sense. (e.g. if $f'(x^*)=0$, then $f''(x^*) > 0 \Rightarrow f(x^*) < f(x) \rightarrow \text{local min.}$)

② Two -Variable Optimization:

consider Taylor series expansion of $f(x, y)$ around (x^*, y^*) :

$$f(x, y) = f(x^*, y^*) + f_x(x^*, y^*) \Delta x^* + f_y(x^*, y^*) \Delta y^* +$$

$$\frac{1}{2} \left\{ f_{xx}(x^*, y^*) (\Delta x^*)^2 + 2 f_{xy}(x^*, y^*) \Delta x^* \Delta y^* + f_{yy}(x^*, y^*) (\Delta y^*)^2 \right\} + O(3).$$

where $f_x = \frac{\partial f}{\partial x}$, $f_y = \frac{\partial f}{\partial y}$, $f_{xx} = \frac{\partial^2 f}{\partial x^2}$, $f_{xy} = \frac{\partial^2 f}{\partial x \partial y}$, $f_{yy} = \frac{\partial^2 f}{\partial y^2}$

ASSUME
 $\Delta x^*, \Delta y^*$
are small

→ This can be re-written in matrix form as follows:

$$f(x, y) = f(x^*, y^*) + [\Delta x^* \quad \Delta y^*] \begin{bmatrix} f_x(x^*, y^*) \\ f_y(x^*, y^*) \end{bmatrix} + \frac{1}{2} [\Delta x^* \quad \Delta y^*] \begin{bmatrix} f_{xx}(x^*, y^*) & f_{xy}(x^*, y^*) \\ f_{xy}(x^*, y^*) & f_{yy}(x^*, y^*) \end{bmatrix} \begin{bmatrix} \Delta x^* \\ \Delta y^* \end{bmatrix}$$

If we denote $\underline{x}^* = \begin{bmatrix} x^* \\ y^* \end{bmatrix}$, $\underline{\Delta x}^* = \begin{bmatrix} \Delta x^* \\ \Delta y^* \end{bmatrix}$, then we can re-write as, $+ O(3)$

$$f(x, y) = f(\underline{x}) = f(\underline{x}^*) + (\underline{\Delta x}^*)^T \underbrace{\nabla f|_{(x^*, y^*)}}_{\text{Gradient (vector)}} + \frac{1}{2} (\underline{\Delta x}^*)^T \underbrace{\nabla^2 f|_{(x^*, y^*)}}_{\text{Hessian (matrix)}} (\underline{\Delta x}^*) + O(3)$$

Gradient = $\nabla f(x, y) = \begin{bmatrix} \partial f / \partial x \\ \partial f / \partial y \end{bmatrix} \rightarrow$ matrix version of 1st derivative.

Hessian = $\nabla^2 f(x, y) = \begin{bmatrix} \frac{\partial^2 f}{\partial x^2} & \frac{\partial^2 f}{\partial x \partial y} \\ \frac{\partial^2 f}{\partial x \partial y} & \frac{\partial^2 f}{\partial y^2} \end{bmatrix} = \underline{H}(\underline{x}^*, y^*)$

• Necessary conditions for the minimum of $f(x, y)$:

$$\nabla f(x^*, y^*) = \begin{bmatrix} f_x(x^*, y^*) \\ f_y(x^*, y^*) \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \end{bmatrix} \Rightarrow \begin{cases} f_x(x^*, y^*) = 0 \\ f_y(x^*, y^*) = 0 \end{cases} \quad \left. \begin{array}{l} \text{i.e 1st derivative} \\ \text{w.r.t each of the} \\ \text{variables is to be zero.} \end{array} \right\}$$

x^*, y^* (two variables) can be found by solving these two equations.

• Sufficient condition for the minimum of $f(x, y)$:

$$\left\{ \frac{1}{2} [\Delta x^* \quad \Delta y^*] \begin{bmatrix} f_{xx}(x^*, y^*) & f_{xy}(x^*, y^*) \\ f_{xy}(x^*, y^*) & f_{yy}(x^*, y^*) \end{bmatrix} \begin{bmatrix} \Delta x^* \\ \Delta y^* \end{bmatrix} \right\} > 0, \text{ for any } (\Delta x^*, \Delta y^*) \text{ (small).}$$

$$\text{i.e } \frac{1}{2} [\Delta x^* \quad \Delta y^*] \underline{H}(\underline{x}^*, y^*) \begin{bmatrix} \Delta x^* \\ \Delta y^* \end{bmatrix} = \frac{1}{2} (\underline{\Delta x}^*)^T \underline{H}(\underline{x}^*, y^*) (\underline{\Delta x}^*) > 0, \quad "$$

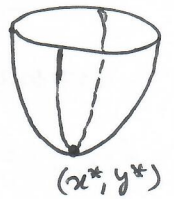
Hessian matrix.

This is called a "Quadratic form"

- For any square matrix $\underline{\underline{A}}$, if $\underline{\underline{v}}^T \underline{\underline{A}} \underline{\underline{v}} > 0$ for any $\underline{\underline{v}}$, then it is said to be positive definite.

For any $(\Delta x^*, \Delta y^*)$

- $\frac{1}{2} (\Delta \underline{\underline{x}}^*)^T \underline{\underline{H}}(x^*, y^*) \Delta \underline{\underline{x}}^* > 0 \Rightarrow$ Positive definite (minimum) $\Rightarrow$ Bottom of a valley



- $\frac{1}{2} (\Delta \underline{\underline{x}}^*)^T \underline{\underline{H}}(x^*, y^*) \Delta \underline{\underline{x}}^* \geq 0 \Rightarrow$ Positive semi-definite (minimum or flat) $\Rightarrow$ A valley fold



- $\frac{1}{2} (\Delta \underline{\underline{x}}^*)^T \underline{\underline{H}}(x^*, y^*) \Delta \underline{\underline{x}}^* < 0 \Rightarrow$ Negative definite (maximum) $\Rightarrow$ peak of a hill



- $\frac{1}{2} (\Delta \underline{\underline{x}}^*)^T \underline{\underline{H}}(x^*, y^*) \Delta \underline{\underline{x}}^* \leq 0 \Rightarrow$ Negative semi-definite $\Rightarrow$ A ridge (maximum or flat)



- $\frac{1}{2} (\Delta \underline{\underline{x}}^*)^T \underline{\underline{H}}(x^*, y^*) \Delta \underline{\underline{x}}^* = 0 \Rightarrow$ Null-definite $\Rightarrow$ just flat (neither minimum nor maximum)



- If $-\infty < \frac{1}{2} (\Delta \underline{\underline{x}}^*)^T \underline{\underline{H}}(x^*, y^*) \Delta \underline{\underline{x}}^* < \infty$, i.e. it is not necessarily positive for all choices of $\Delta \underline{\underline{x}}^*$, or always negative for all choices of $\Delta \underline{\underline{x}}^*$.

(for some choices of $\Delta \underline{\underline{x}}^*$, it is positive & for some other it is negative),

THEN $\underline{\underline{H}}$ is said to be indefinite (when evaluated at (x^*, y^*))

and (x^*, y^*) is called a saddle point.

(can have minimum along one direction & maximum w.r.t some other direction).



③ Multi-Variable Optimization:

$f(\underline{\underline{x}}) =$

Consider a function of n variables, $f(x_1, x_2, \dots, x_n)$; $\underline{\underline{x}} = \begin{Bmatrix} x_1 \\ x_2 \\ \vdots \\ x_n \end{Bmatrix}$

Taylor's series expansion is given by, (about $\underline{\underline{x}}^*$)

$$\begin{aligned} f(\underline{\underline{x}}) &= f(\underline{\underline{x}}^*) + \sum_{i=1}^n \left. \frac{\partial f(\underline{\underline{x}})}{\partial x_i} \right|_{\underline{\underline{x}}=\underline{\underline{x}}^*} \cdot (x_i - x_i^*) + \frac{1}{2} \sum_{i=1}^n \sum_{j=1}^n \left. \frac{\partial^2 f(\underline{\underline{x}})}{\partial x_i \partial x_j} \right|_{\underline{\underline{x}}=\underline{\underline{x}}^*} \cdot (x_i - x_i^*)(x_j - x_j^*) + \mathcal{O}(3) \\ &= f(\underline{\underline{x}}^*) + (\nabla f(\underline{\underline{x}}^*))^T \Delta \underline{\underline{x}}^* + \frac{1}{2} (\Delta \underline{\underline{x}}^*)^T \underline{\underline{H}}(\underline{\underline{x}}^*) \Delta \underline{\underline{x}}^* + \mathcal{O}(3) \end{aligned}$$

- Necessary condition for $\underline{x}^*$ to be an extremum (a stationary point)
 $\hookrightarrow$ minimum (or) maximum

Gradient of an n -variable function $\nabla f(\underline{x}^*) = \begin{Bmatrix} \frac{\partial f}{\partial x_1} \big|_{\underline{x}^*} \\ \frac{\partial f}{\partial x_2} \big|_{\underline{x}^*} \\ \vdots \\ \frac{\partial f}{\partial x_n} \big|_{\underline{x}^*} \end{Bmatrix} = \begin{Bmatrix} 0 \\ 0 \\ \vdots \\ 0 \end{Bmatrix}$ $\rightarrow n$ equations.

n variables ($x_1^*, x_2^*, \dots, x_n^*$);

- Sufficient condition: $\underline{H}$ @ (x^*, y^*) should be positive definite for a minimum of $f(\underline{x})$.

Hessian of an n -variable function. $\underline{H}(\underline{x}^*) = \begin{Bmatrix} \frac{\partial^2 f}{\partial x_1^2} \big|_{\underline{x}^*} & \frac{\partial^2 f}{\partial x_1 \partial x_2} \big|_{\underline{x}^*} & \dots & \frac{\partial^2 f}{\partial x_1 \partial x_n} \big|_{\underline{x}^*} \\ \frac{\partial^2 f}{\partial x_2 \partial x_1} \big|_{\underline{x}^*} & \frac{\partial^2 f}{\partial x_2^2} \big|_{\underline{x}^*} & \dots & \dots \\ \vdots & \vdots & \ddots & \vdots \\ \frac{\partial^2 f}{\partial x_n \partial x_1} \big|_{\underline{x}^*} & \dots & \dots & \frac{\partial^2 f}{\partial x_n^2} \big|_{\underline{x}^*} \end{Bmatrix}$

$\Delta \underline{x}^{*T} \underline{H}(\underline{x}^*) \Delta \underline{x}^* > 0$ for all $\Delta \underline{x}^*$.
 for $f(\underline{x}^*)$ to be minimum.

- A matrix $\underline{A}$ will be positive definite if all its eigenvalues are positive; that is, all the values of λ that satisfy the determinantal equation $|\underline{A} - \lambda \underline{I}| = 0$ should be positive.
 $\hookrightarrow$ Identity matrix
- Similarly, $\underline{A}$ will be negative definite if its eigenvalues are negative.
- Consider the following "n" determinants of a matrix $\underline{A}_{n \times n}$:

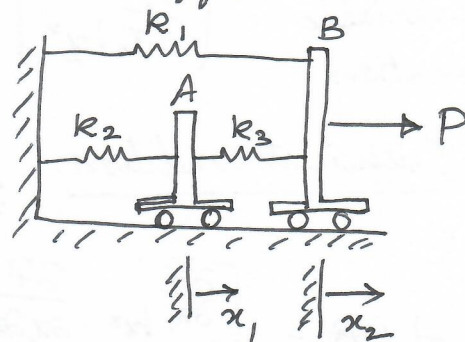
$$A_1 = |a_{11}|, A_2 = \begin{vmatrix} a_{11} & a_{12} \\ a_{21} & a_{22} \end{vmatrix}, A_3 = \begin{vmatrix} a_{11} & a_{12} & a_{13} \\ a_{21} & a_{22} & a_{23} \\ a_{31} & a_{32} & a_{33} \end{vmatrix}, \dots, A_n = \begin{vmatrix} a_{11} & a_{12} & \dots & a_{1n} \\ a_{21} & a_{22} & \dots & \vdots \\ \vdots & \vdots & \ddots & \vdots \\ a_{n1} & \dots & \dots & a_{nn} \end{vmatrix}$$

Then, matrix $\underline{A}$ will be positive definite if and only if all the values $A_1, A_2, \dots, A_n$ are positive. Matrix $\underline{A}$ will be negative definite if and only if the sign of A_j is $(-1)^j$ for $j = 1, 2, \dots, n$. If some of the A_j are positive and the remaining A_j are zero, then the matrix $\underline{A}$ will be positive semidefinite.

④ Examples

Problem
(i) Consider two frictionless rigid bodies (carts) A and B connected by three linear elastic springs having spring constants k_1, k_2 , and k_3 . The springs are at their natural positions when the applied force P is zero. Find the displacements x_1 and x_2 under the force P by using the principle of minimum potential energy.

Solution: According to the principle of minimum potential energy, the system will be in equilibrium under the load P if the potential energy is a minimum. The potential energy of the system is given by,



— work done by external forces

$$\Rightarrow U = \left[\frac{1}{2} k_2 x_1^2 + \frac{1}{2} k_3 (x_2 - x_1)^2 + \frac{1}{2} k_1 x_2^2 \right] - P x_2$$

We need to find the x_1, x_2 corresponding to equilibrium i.e. minimize $U(x_1, x_2)$.

Necessary conditions for the minimum of U are

$$\frac{\partial U}{\partial x_1} = k_2 x_1 - k_3 (x_2 - x_1) = 0 \quad \& \quad \frac{\partial U}{\partial x_2} = k_3 (x_2 - x_1) + k_1 x_2 - P = 0$$

The stationary (or extreme) points x_1^*, x_2^* are obtained by solving these two equations:

$$x_1^* = \frac{P k_3}{k_1 k_2 + k_1 k_3 + k_2 k_3} \quad ; \quad x_2^* = \frac{P (k_2 + k_3)}{k_1 k_2 + k_1 k_3 + k_2 k_3}$$

Sufficiency conditions for the minimum at (x_1^*, x_2^*) can be verified by testing the positive definiteness of the Hessian matrix of U .

$$H(x_1^*, x_2^*) = \begin{bmatrix} \left. \frac{\partial^2 U}{\partial x_1^2} \right|_{\underline{x}^*} & \left. \frac{\partial^2 U}{\partial x_1 \partial x_2} \right|_{\underline{x}^*} \\ \left. \frac{\partial^2 U}{\partial x_1 \partial x_2} \right|_{\underline{x}^*} & \left. \frac{\partial^2 U}{\partial x_2^2} \right|_{\underline{x}^*} \end{bmatrix} = \begin{bmatrix} k_2 + k_3 & -k_3 \\ -k_3 & k_1 + k_3 \end{bmatrix}$$

let us find the determinants H_1 & H_2 .

$$H_1 = |K_2 + K_3| = K_2 + K_3 > 0$$

$$H_2 = \begin{vmatrix} K_2 + K_3 & -K_3 \\ -K_3 & K_1 + K_3 \end{vmatrix} = K_1 K_2 + K_1 K_3 + K_2 K_3 > 0$$

Since the spring constants are always positive, $H_2(x_1^*, x_2^*)$ is indeed positive definite. Hence, x_1^*, x_2^* correspond to the minimum potential energy, and therefore ^{are} the equilibrium displacements.

Problem (ii) Find the extreme points of the following function and their nature.

$$f(x_1, x_2) = x_1^3 + x_2^3 + 2x_1^2 + 4x_2^2 + 6$$

Exercise: Make a surface plot of this function in MATLAB

Solution: Necessary conditions for the existence of an extreme point are

$$\frac{\partial f}{\partial x_1} = 0 \Rightarrow 3x_1^2 + 4x_1 = x_1(3x_1 + 4) = 0 \Rightarrow x_1 = 0 \text{ or } -4/3$$

$$\frac{\partial f}{\partial x_2} = 0 \Rightarrow 3x_2^2 + 8x_2 = x_2(3x_2 + 8) = 0 \Rightarrow x_2 = 0 \text{ or } -8/3$$

Hence, we get four stationary points $(x_1^*, x_2^*) : \{(0, 0), (0, -8/3), (-4/3, 0), (-4/3, -8/3)\}$

To find the nature of these extreme points, we have to use the sufficiency conditions. The second-order partial derivatives of f are:

$$\frac{\partial^2 f}{\partial x_1^2} = 6x_1 + 4; \quad \frac{\partial^2 f}{\partial x_2^2} = 6x_2 + 8; \quad \frac{\partial^2 f}{\partial x_1 \partial x_2} = 0$$

$$\text{Hessian matrix, } H(x_1, x_2) = \begin{bmatrix} 6x_1 + 4 & 0 \\ 0 & 6x_2 + 8 \end{bmatrix}$$

$$\text{Determinants are: } H_1 = |6x_1 + 4| = 6x_1 + 4 \text{ and } H_2 = \begin{vmatrix} 6x_1 + 4 & 0 \\ 0 & 6x_2 + 8 \end{vmatrix} = (6x_1 + 4)(6x_2 + 8)$$

Point $\underline{x}$	H_1 value	H_2 value	Nature of H_2	Nature of $\underline{x}$	$f(\underline{x})$
$(0, 0)$	+4	+32	Positive definite	Local minimum	6
$(0, -8/3)$	+4	-32	Indefinite	Saddle point	418/27
$(-4/3, 0)$	-4	-32	Indefinite	Saddle point	194/27
$(-4/3, -8/3)$	-4	+32	Negative definite	Local maximum	50/3

Recall: matrix A is pos-def if all A_{jj} are positive
 " " neg-def iff sign of A_{jj} is $(-1)^j$
 $\Rightarrow$ for $A_{22} \rightarrow +ve$, $A_{11} \rightarrow -ve$.